

Review

① Independence of multiple events $\rightarrow$

② Basics of counting.

permutations

Divided into r classes, then

$$n = n_1 \times n_2 \times \dots \times n_r$$

$A_1, A_2, \dots, A_n$

are independent, if

$$P\left(\bigcap_{i \in S} A_i\right) = \prod_{i \in S} P(A_i)$$

Logistics

Quiz day

Wed: 9-10

Permutations: # of ways to pick k distinct objects from n distinct objects and ordering them or arranging them

Combinations # of ways to pick k distinct objects out of n distinct objects

19. Proving that

$\mathbb{C}$ is uncountable.

$\downarrow$ them
 $1, 2, \dots, n$
positions
 $(P_k) = n(n-1) \dots (n-k+1)$

Three Prisoners Problem

Suppose # of ways to pick k -distinct objects = A

of ways to arrange = $k!$

$$\rightarrow A \cdot k! = {}^n P_k \Rightarrow A = \frac{{}^n P_k}{k!} \cong {}^n C_k = \frac{n!}{(n-k)! k!}$$

1, 2, ..., k from 1, 2, ..., n
of ways to arrange k objects = $k \times (k-1) \times \dots \times 1$ ways = $k!$

of k -element subsets of $n = \binom{n}{k}$

Combinatorial identities

① # of possible subsets of $\{1, 2, \dots, n\}$
 $= 2^n$

$$= \sum_{k=0}^n \{\text{\# of subsets of } k\}$$

$$2^n = \sum_{k=0}^n \binom{n}{k}$$

$$\textcircled{2} \quad \binom{n}{k} = \binom{n-1}{k-1} + \binom{n-1}{k}$$

#

$$\binom{n}{k}$$

$$\{1, 2, 3\}$$

$$\{1, 2\}$$

Counting



of k -element ^{Subsets} of $[n]$

= # of k -element subsets of $[n]$ containing 1
+ # of k -element subsets of $[n]$ without 1

$$\binom{n}{k} = \binom{n-1}{k-1} + \binom{n-1}{k}$$

$n=5, k=2$
 $\{1,2\}, \{1,3\}, \{1,4\}, \{1,5\}$

$n=5, k=2$
 $\{2,3\}, \{2,4\}, \{2,5\}$

$\{1,2,3\}$
 $\{1,2,3\}$

$$n \cdot 2^{n-1} = \sum_{k=0}^{n-1} n \binom{n-1}{k}$$
$$= \sum_{k=1}^n \binom{n}{k} k$$

Counting # of subsets with a designated leader / CR.

$$\text{LHS} = 1 \times \underbrace{2 \times 2 \times \dots \times 2}_{n-1 \text{ times}}$$

first

RHS: select group of size $k+1$ where $k \in \{0, \dots, n-1\}$

$$\binom{n}{1} \binom{n-1}{k}$$

selecting
the leader

$$\Rightarrow n \cdot \sum_{k=0}^{n-1} \binom{n-1}{k} = 2^{n-1} n$$

Second RHS: select group of k people $k \in \{1, 2, \dots, n\}$
 & then choose a leader

$$\rightarrow \binom{n}{k} \times k \Rightarrow \sum_{k=1}^n k \binom{n}{k} = n \cdot 2^{n-1}$$

Number of permutations possible for the word LETTER

$$LETTER = L E T T E R$$

$$6!$$

$$\frac{6!}{2!2!}$$

of non-negative solns to $x_1 + x_2 + \dots + x_r = n$

$$\boxed{\binom{n+r-1}{r-1}}$$

$$r \geq 3$$

for any $n \geq 0$

$$x_1 + x_2 + \dots + x_r = n$$

$n+r-1$ dots

of vectors

$(x_1, x_2, \dots, x_r)$ of lengths

$$r \text{ st } \sum_{i=1}^r x_i \leq n \quad x_i \text{'s non-negative}$$

$$\sum_{i=1}^r x_i = j \quad \forall j \leq n$$

$$\sum_{j=0}^n \binom{n-j}{r-1}$$

Partitioning of set of n containing n distinct objects into r subsets

of sizes $n_1, n_2, \dots, n_r$ element $\sum_{i=1}^r n_i = n$

$$\binom{n}{n_1} \binom{n-n_1}{n_2} \dots \binom{n-n_1-\dots-n_{r-1}}{n_r}$$

$$= \frac{n!}{\cancel{(n-n_1)!} n_1!} \times \frac{\cancel{(n-n_1-n_2)!}}{\cancel{(n-n_1-n_2)!} n_2!} \dots \frac{n_r!}{\cancel{0!} n_r!}$$

$$= \frac{n!}{n_1! n_2! \dots n_r!}$$


$$n = \{1, 2, 3, 4\} \quad n_1 = 2, n_2 = 2$$

$$\{1, 2\} \quad \{3, 4\}$$

+

$$\{3, 4\} \quad \{1, 2\}$$

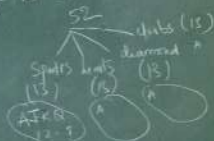
The order in which these subsets are arranged matters in this count.

Examples

- ① Find # of ways to distribute deck of 52 cards among 4 people?

Dividing 52 into ^{four} subsets of size 13 each

$$\frac{52!}{(13!)^4}$$



Find # of ways to distribute such ^{that} each person gets an ace

of ways $\left(4!\right) \times \left(\frac{48!}{(12!)^4}\right)$

Distribute aces

of ways to distribute rest of the cards

of ways to distribute these cards st exactly one person gets all 13 spades =

$$\binom{4}{1} \times \frac{39!}{(13!)^3}$$

of ways to
choose the
person who
gets all the
Spades

of ways
to distribute
across the rest
of the three

Three Prisoners Problem

T : Tintin is sentenced

H : Haddock "

C : Calculus "

$$\Omega = \{ \{YNN\}, \{NYN\}, \{NNY\} \}$$

$$T = \{YNN\}, H = \{NYN\}$$

$$C = \{NNY\}$$

$$P(T) = P(H) = P(C) = \frac{1}{3}$$

Guard can either say Haddock is pardoned
or Calculus is Pardoned

G_C = Grand said that
Calculus is pardoned = $\{ \{YNN\}, \{NTN\} \}$

$$P(T | G_C) = \frac{P(G_C | T) P(T)}{P(G_C)}$$

$$P(T^c | G_C) = \frac{\frac{1}{2} \times \frac{1}{3}}{\frac{1}{6}} = \frac{1}{3}$$

Ref: Data Science by Stanley Chan

$$\Rightarrow P(G_C | T) = \frac{1}{2}, P(G_C | C) = 0, P(G_C | H) = 1$$
$$P(G_C) = \frac{1}{2} \times \frac{1}{3} + 1 \times \frac{1}{3} = \frac{1}{3} \times \frac{3}{2} = \frac{1}{2}$$

$$P(G_c | I^c) = \frac{P(G_c \cap T^c)}{P(T^c)}$$

$$= \frac{(1/3)}{(2/3)} = \frac{1}{2}$$

Note

T^c, H^c, C^c are not disjoint. Can't use total probability theorem w.r.t those events.

Cantor Set

$$G_0 = \left[0, 1 \right]$$

$$G_1 = \left[0, \frac{1}{3} \right] \cup \left[\frac{2}{3}, 1 \right]$$

$$G_2 = \left[0, \frac{1}{9} \right] \cup \left[\frac{2}{9}, \frac{1}{3} \right] \cup \left[\frac{2}{3}, \frac{7}{9} \right] \cup \left[\frac{8}{9}, 1 \right]$$

Similarly $G_c = \{YNN, NYN\}$

$$P(G_c | C^c) = \frac{P(G_c \cap X^c)}{P(X^c)}$$

$$P(G_c | H^c) = \frac{1}{2/3} = \frac{1}{2}$$

$$I = [0, 1]$$

$$P((a, b)) = b - a, 0 \leq a \leq b \leq 1$$



$$P(Q \cap [0, 1]) = 0 \quad \begin{matrix} \text{A countable} \\ \text{set} \end{matrix} \quad \begin{matrix} A \subseteq [0, 1] \\ \rightarrow P(A) = 0 \end{matrix}$$

$$C = \bigcap_{n=1}^{\infty} C_n$$

$$P(C) = P\left(\bigcap_{n=1}^{\infty} C_n\right)$$

$$P(C_n) = \left(\frac{2}{3}\right)^n$$

Continuity property Corollary $\leftarrow \lim_{n \rightarrow \infty} P(C_n)$

$$= \lim_{n \rightarrow \infty} \left(\frac{2}{3}\right)^n = 0$$

0.2020202...

$$C \leftrightarrow \{0, 2\}^{\mathbb{N}}$$

$$0 \times \left(\frac{1}{3}\right) + 2 \times \left(\frac{1}{3}\right)^2 + 2 \times \left(\frac{1}{3}\right)^3 \leftarrow 0.22022 \dots$$

$$\sum_{i=1}^{\infty} a_i \left(\frac{1}{3}\right)^i \leftarrow a_1 a_2 a_3 \dots a_i \in \{0, 2\}$$

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$$\{0, 2\}^* \rightarrow \mathbb{C}$$

$$a_1 a_2 a_3 \dots \rightarrow a = \sum_{i=1}^{\infty} a_i \left(\frac{1}{3}\right)^i$$

$$\left. \begin{array}{l} \text{if } a_1 = 0 \quad a \in \left[0, \frac{1}{3}\right] \\ \text{otherwise } a \in \left[\frac{2}{3}, 1\right] \\ \text{if } a_2 = 0 \\ \quad 3\left(a - \frac{a_1}{3}\right) \in \left[0, \frac{1}{3}\right] \\ \text{otherwise } \in \left[\frac{2}{3}, 1\right] \end{array} \right\} \Rightarrow a \in \mathbb{C}$$

$$\mathbb{C} \rightarrow \{0, 2\}^*$$

$$x \rightarrow x_1 x_2 x_3 \dots$$

$$i=1.$$

while ($x \neq 0$)

$$\text{if } x \geq \frac{1}{3}$$

$$x_i = 2.$$

otherwise

$$x_i = 0.$$

$$x = 3\left(x - \frac{x_i}{3}\right)$$

$$\frac{1}{3} \rightarrow 0222\dots$$

$$\frac{2}{3} \rightarrow 2$$

Since we know $\{0, 2\}^*$ is uncountable
 $\Rightarrow \mathbb{C}$ is uncountable